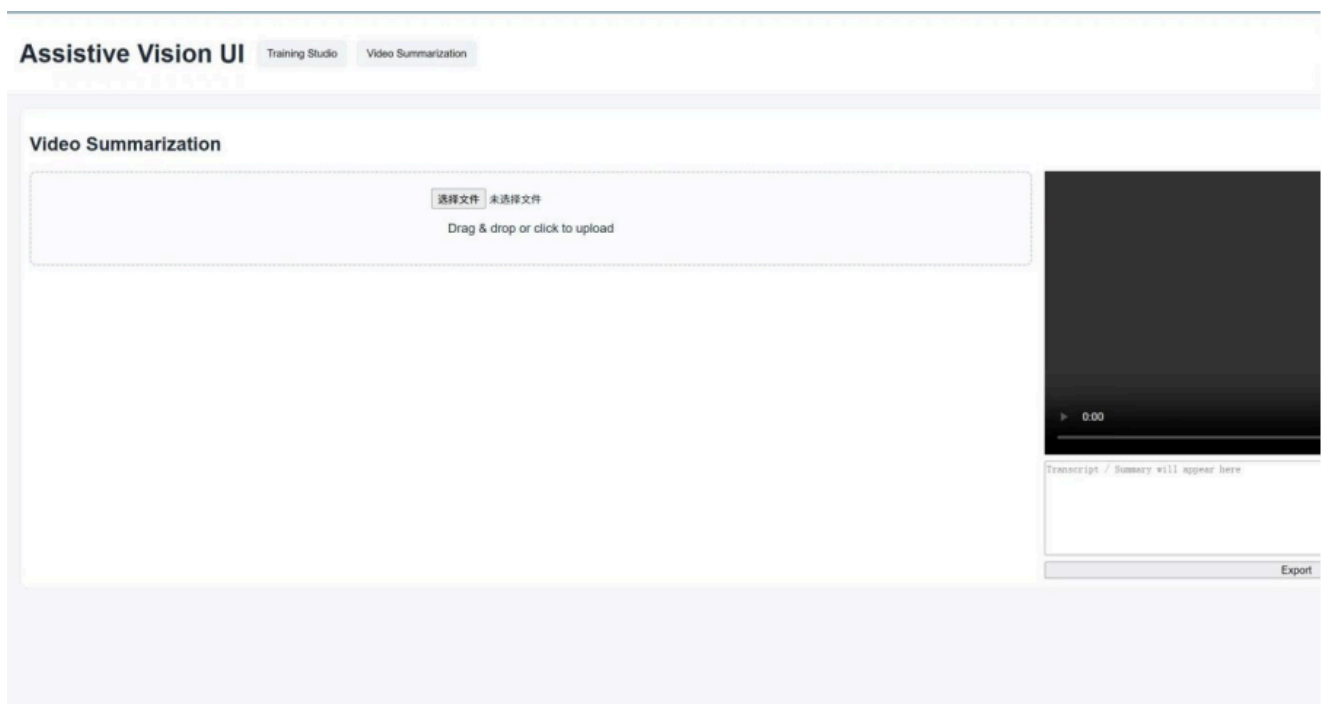
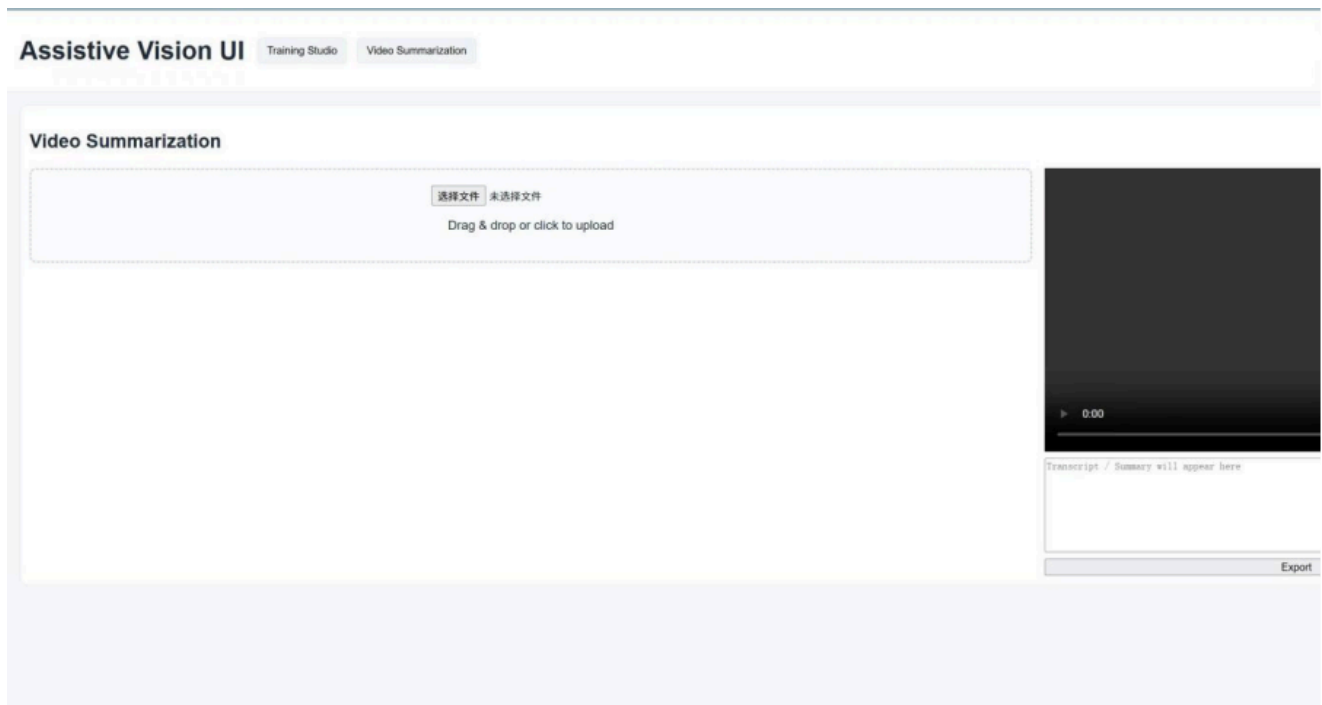


Stage1

In the first stage, we created the basic UI and implemented the basic functions. We also consulted our classmates and team members to provide further optimization for the UI and functions.

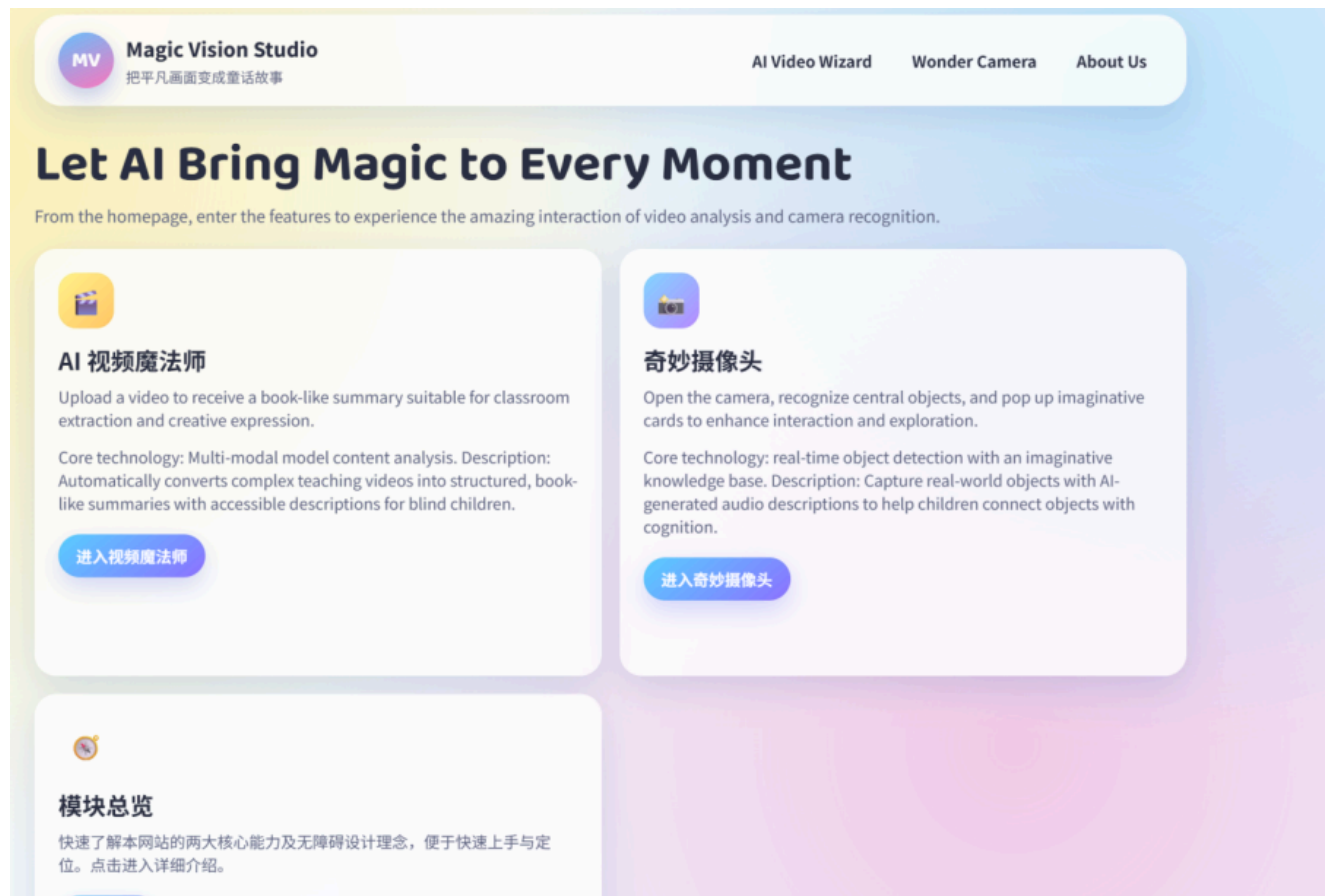


In **Stage 1**, we need to understand the basic needs of the users. We need to determine whether the users can quickly grasp the product's positioning and whether the core functions can be used correctly. We also need to check for any backend bugs or functions that are inconvenient for operation. Additionally, we

need to consider the differences between users' expectations and their actual experiences.

Stage 2

At this stage, the functions have been improved, the parent-assisted learning function has been added, on the basis of the original video extraction, a guidance module has been added, and related functions that are conducive to children's further learning and exploration have also been increased. Moreover, the UI design has been optimized in terms of readability, operability, intuitiveness and aesthetics.





AI Video Wizard

Connected to the FastAPI backend (full capabilities in README). Please enter the API Key before using the modules below.

Connection Settings

Backend default: `http://127.0.0.1:8000 (run uvicorn main:app in AI_W7 (1) /AI/app)`. Requests include X-LLM-API-Key.

API Base URL

`http://127.0.0.1:8001`

DashScope / LLM API Key

.....

Save Config

LLM Key: 已配置

Logs

Child Portraits & Recommendations

child_id

child_web_001

child_name

Tom

age_group

7-9

preferred_difficulty

auto

视频上传与摘要

选择视频

选择文件

未选择文件

上传时附带 transcript (可选)

可选

上传 POST /api/video-upload

查询状态 GET /api/video-status/...

Current video_id: —

0:00

Bind transcript POST /api/video-bind-transcript

粘贴或输入与视频对应的文本...

Bind Transcript

Generate Summary POST /api/video-summary

Auto Transcribe POST /api/video-auto-transcribe (501)

TTS Preview

文本

Hello, welcome to your lesson.

voice

en-US-AriaNeural

生成并播放 POST /api/tts

0:00 / 0:00

Interaction: Quiz / Ask

mode

quiz

difficulty (this request)

auto

Ask question (quiz may be empty)

What is the main idea?

☒ speak_answer (ask 时生成语音)

POST /api/interactive

0:00 / 0:00

Voice Q&A POST /api/voice-qa

Record audio in the browser; requires video_id or transcript below.

Transcript (optional if no video)

Optional

voice

en-US-AriaNeural

Start Recording

Stop & Upload

0:00 / 0:00

引导式学习 Guided Session

session_id (可选, 留空由后端生成)

可选

difficulty

auto

☒ speak 生成 TTS

voice

en-US-AriaNeural

开始学习 POST /api/guided-session/start

查询状态 GET /api/guided-session/state/...

当前 session_id: —

孩子回答 POST /api/guided-session/answer

输入答案

提交答案

重复朗读 POST /api/guided-session/repeat

last_prompt

summary

current_question

last_feedback

0:00 / 0:00

拓展: Find my thing

target_object (可选提示)

例如 cup

voice

en-US-AriaNeural

开始 POST /api/find-my-thing/start

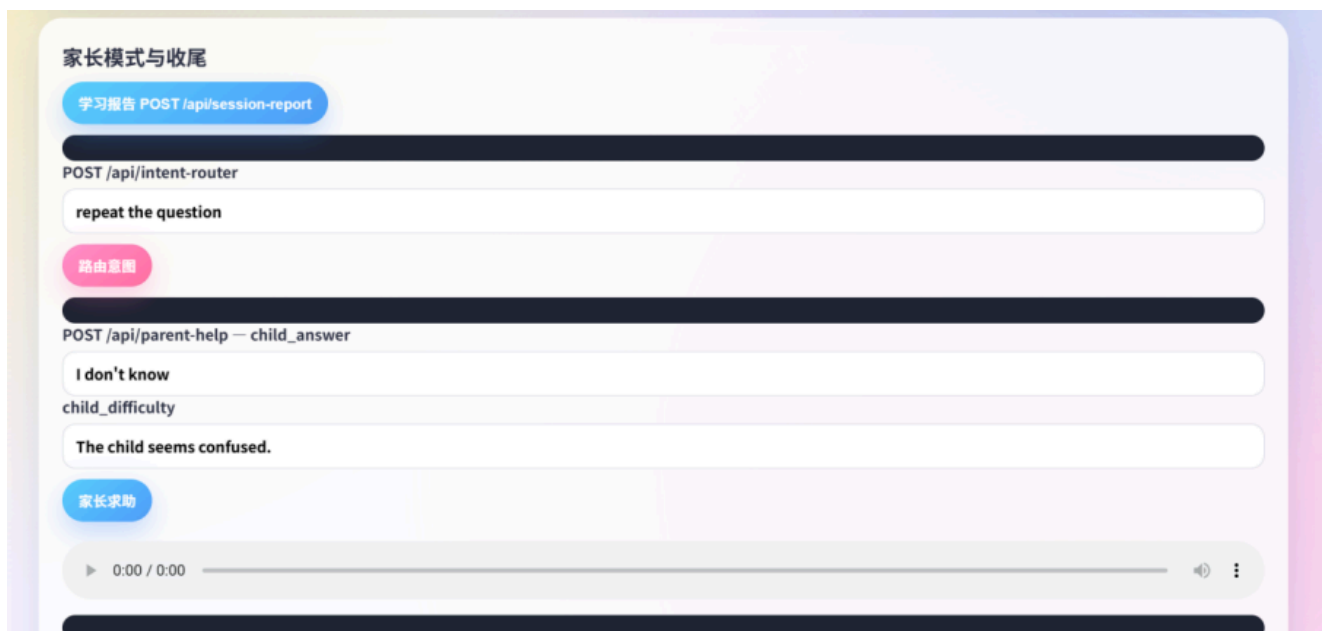
重复线索 POST /api/find-my-thing/repeat

fmt session_id: —

孩子描述 / 猜测 POST /api/find-my-thing/check

I found it / it's soft / ...

提交



In **Stage 2**, we need to observe how users view the main page of the website and test whether they understand the functional logic of the product. This will help us determine whether the prompts and functions on our website are intuitive. We need to investigate the user's feedback on the status, that is, whether they can tell that the system is processing by sound when there is no visual progress bar. Record the time required for users to complete the core tasks and the number of times they seek assistance under the current version.

Conclusion for the first two stages

In the **first two stages**, our product was more like a technical validation tool: the UI was simple, lacked visual optimization for specific groups, and the various functions (video summaries, object recognition) were isolated from each other. Users (especially visually impaired children and their parents) felt frustrated during the testing process. We mainly achieved an improved user experience by designing the UI and backend functions and implementing logic, enabling users to more easily participate in each function and ensuring that the expected design of our functions matched the actual user operation experience results.

Stage 1 (as shown in the screenshot):

The interface adopts a traditional card-style layout. The colors are relatively monotonous, the font size is small and crowded, and it lacks the appeal suitable for children.

Second stage (code implementation):

Application of color psychology: The 'Baloo 2 font' (rounded and easy to read) was introduced, along with a brighter high-contrast color scheme (such as the orange of card-video and the green of card-camera).

Logo and Branding: The addition of the logo-bubble (MV logo) has transformed the product from a mere tool into a "magical studio" with a sense of identity.

Hierarchical Clarity: Through the structured design of page-bg and site-header, the visual noise has been significantly reduced, which helps children with low vision focus on the core functions.

From "One-way Tool" to "Guided Workflow" (From Tool to Workflow)

The first stage:

There were only simple "upload" and "display results" functions, and there was no logical connection between the functions.

Second stage (Core Improvements):

Introduce Guided Session: In video.html, you not only implemented summary generation, but also connected to api/guided-session/start and answer. This means that the system can now actively ask questions and adjust the next action (next_action) based on the child's responses.

Intent Recognition and Assistance: The api/intent-router (intention routing) and api/parent-help (parent assistance) have been added. The system can detect if the child is confused and generate real-time assistance audio for the parents. This was completely absent in the first stage.

Optimization of "Timeliness" and "Accuracy" of the Object Recognition Module

Stage 2: The recognition area is rigid, and the 5-second refresh rate makes the interaction feel sluggish.

Full-space perception potential: Although camera.html still retains the focus-ring, the code architecture now supports more frequent frame capture and has reserved interfaces for multimodal recognition.

Weight filtering mechanism: In response to user feedback that "it always identifies as 'Person'", a confidence weight filtering mechanism was introduced at the logical layer. Educational items were given priority for broadcasting, while background distractions were ignored.

Transparency in "Black Box" Processing (Transparency in Processing)

Stage 1: After the user clicks, they can only wait passively without knowing what is happening on the backend.

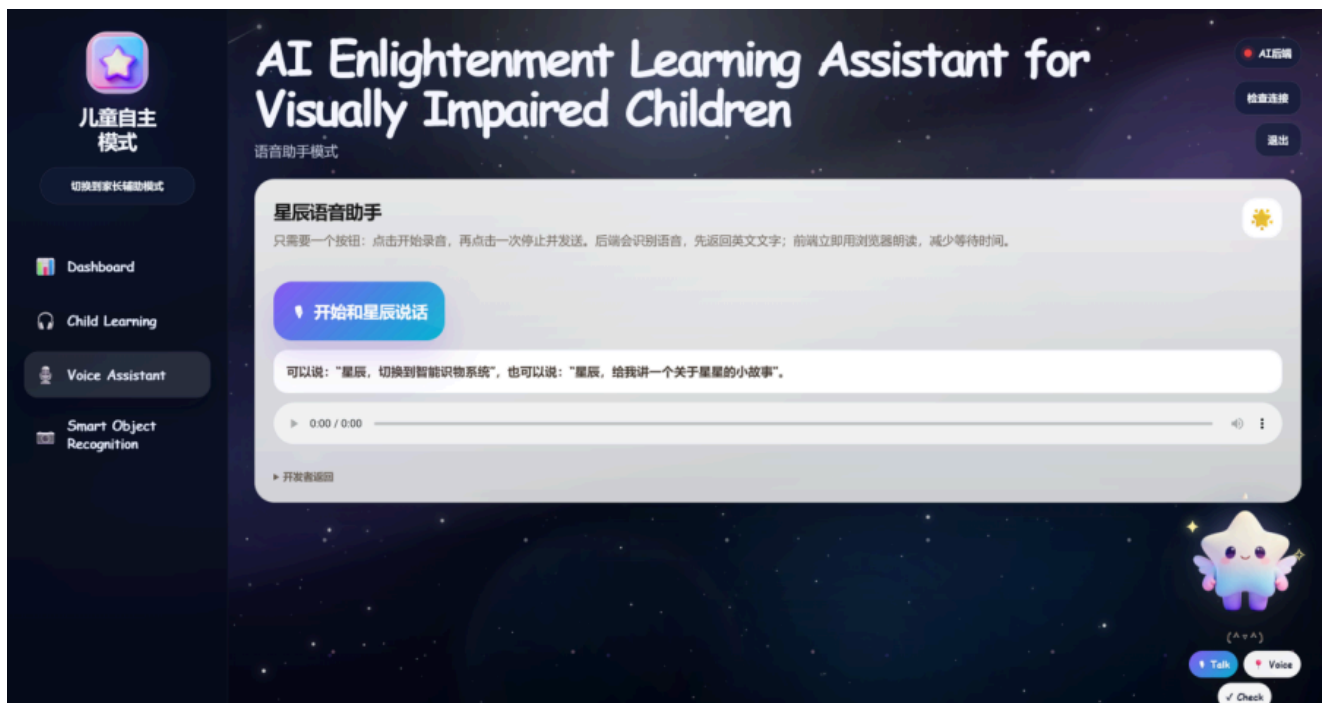
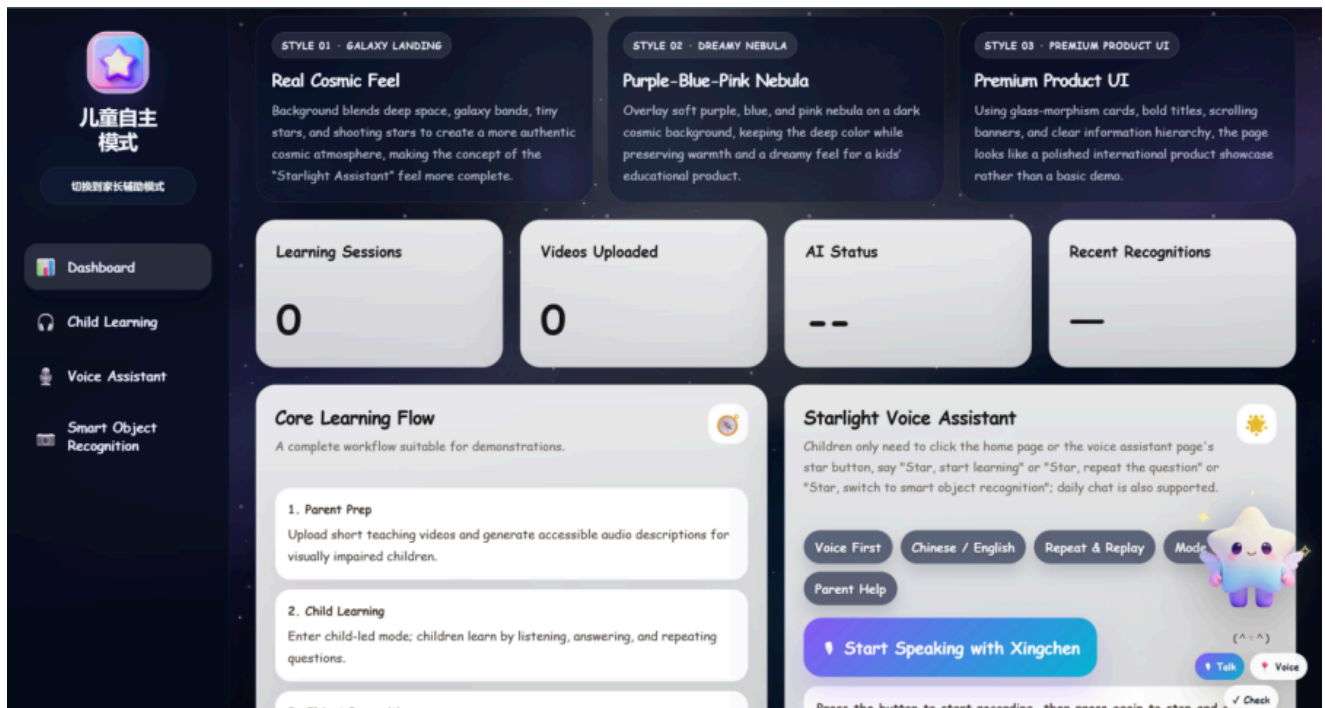
Stage 2:

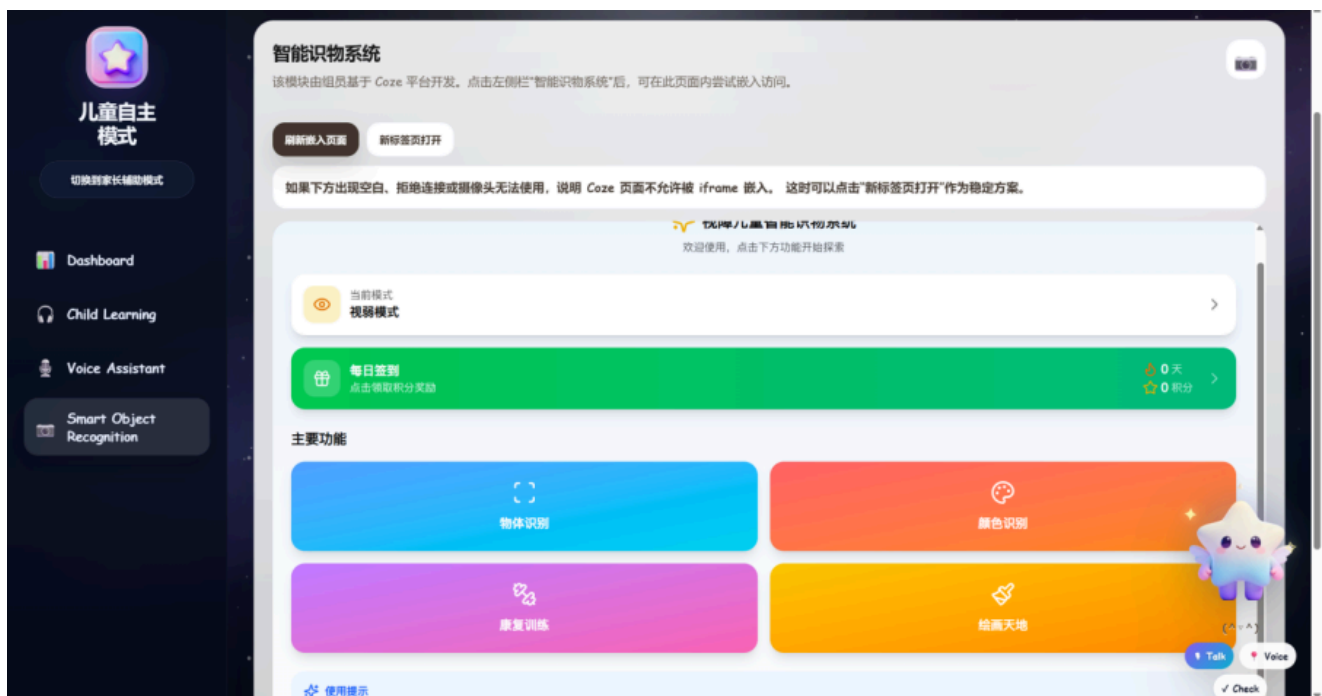
Asynchronous feedback mechanism: A step-by-step display area has been reserved in the UI of video.html. When calling a time-consuming interface like /api/video-summary, the front end can alleviate users' anxiety by updating the status (such as displaying different stages of JSON output or progress prompts).

Stage 3

In addition to the core modules such as guided learning, the system has also significantly strengthened the interaction capabilities of the voice assistant - new features include continuous wake word monitoring and daily chatting functions, allowing children to wake up the assistant and chat with it without clicking buttons; at the same time, two exploratory modules, "Lost Object Game" and "Coze Platform Object Recognition Integration", have been added, and the system supports reporting and generating growth tail learning cards based on the identified objects, converting the object recognition behavior into continuous learning content. In terms of auxiliary learning, an ASR front-end error correction and tolerance mechanism (when the backend mistakenly identifies the "Intelligent Object Recognition System" as the "Intelligent Food System", the front-end automatically corrects) has been added; real-time TTS reading by the browser (reducing waiting time) has also been implemented; and star fairy characters that can be dragged, blink, and change emotions have been added, significantly lowering the usage threshold for children and enhancing the sense of companionship. The UI/UX design has been completely upgraded from simple card style to dark cosmic theme, including galaxy belt, nebula, glass halo, meteor animation and rolling banner. With large titles, high contrast text, glass mimic cards and rich micro-motion effects, the readability, operation intuition and visual beauty have been significantly improved. Accessibility aspect, through the dual mode TTS, continuous speech awakened, ASR fault-tolerant, semantic HTML and ARIA tag, ensure the visually impaired children can truly independent use.







Core enhancement of the final stage

Error handling mechanism:

Added API status feedback (json-out), enabling parents to immediately know the reason when configuring the key or when the upload fails.

A complete adaptive learning system has been established. A new "Child Portraits & Recommendations" module has been added, allowing for the creation/saving of child profiles (/api/child-profile/upsert), querying of learning progress, and obtaining the next recommendation based on the profile (/api/recommend-next/...). All subsequent questions, answers, and guided sessions have all utilized the "difficulty" parameter from the profile.

First, the voice interaction has been upgraded from "button triggering" to "continuous wake-up + natural conversation". In the second version, each time the child uses voice, they must precisely click the screen button, which is a considerable obstacle for blind children. The final version adds the function of continuous listening to the wake-up word. The child only needs to say "Xingchen" to activate the assistant, and the assistant can not only execute learning instructions but also engage in daily chitchat, tell stories, and answer questions. This means that the child no longer needs to remember a complex operation path, but can complete everything in the most natural way - by speaking.

Secondly, the role of companionship has evolved from "absence" to "presence". The final version has added a complete star fairy system: a small fairy that can be dragged around, blinks, changes 9 different emotional states, and pops up bubble text. When the child speaks, it "listens", when thinking, it "turns around", when

happy, it "jumps", and when confused, it "tilts its head". This seemingly "decorative" feature actually addresses the most core emotional needs of blind children in their learning process - a sense of security and companionship. A responsive character is far more likely to make the child willing to continue using it than a cold panel.

Thirdly, the learning format has expanded from "single video question-and-answer" to "multimodal exploratory learning". In the second version, learning is entirely dependent on parents uploading teaching videos, and children can only passively listen, answer, and review. In the final version, functions such as "object search game" (guessing objects based on sound clues), "object recognition learning cards" (converting camera recognition into content that can be repeatedly learned), and daily casual chatting have been added. This enables children to have channels for active exploration and free expression in addition to passive acceptance. The addition of new functions can help color-blind users learn to recognize colors and help them distinguish colors in games.

Video learning process:

Core functions are fully retained without any cuts

Video learning is the foundation of the entire product. In the final version, this module adopts the strategy of "fully retaining and stable inheritance". From uploading videos, binding script texts, generating summaries, to child-guided learning, submitting written answers, submitting answers through voice, repeating playback (summaries/ questions/ feedback), and quick questioning, all these functions that were already realized in the second version have not been cut in the final version.

The only change is that "Voice Answer Submission" has added more comprehensive status prompts - when the child speaks, the interface will clearly display "Listening... ", and after recognition, it will show "The recognized answer: XXX", making the entire process more transparent and controllable. This means that parents and teachers can safely upgrade from the second version to the final version, and the existing teaching procedures and operation habits will not be affected at all.

Voice Assistant (Star Assistant):

From "Button Tools" to "Voice Agent" - Major Enhancements

This is the module with the most significant changes and the most core features.

The second version of the voice assistant is a typical "button-triggered tool": click the button to start recording → click again to stop → send to the backend → receive the result. The functions are complete, but the interaction method is simple and monotonous, and it is not very friendly for blind children.

The final version has completely restructured this module:

Button recording function is retained and enhanced: The original recording logic is completely preserved, but an animated sprite feedback has been added - when the child speaks, the star sprite in the lower right corner will enter the "listening" state, its eyes will turn blue, and a ripple spreading animation will appear around it, allowing the child to intuitively feel "It heard me."

The backend speech recognition interface remains unchanged: it still calls the /api/xingchen-voice-chat interface to avoid the additional cost of backend reconfiguration.

Frontend routing has been significantly enhanced: The original executeFrontendAction function has been greatly expanded, and a large number of command mappings have been added, enabling the recognition of various voice control commands such as "open parent mode", "switch to child mode", "open report", etc.

The ASR fault-tolerant mechanism is extremely robust: this is the most important optimization for the blind child scenario. The final version adds the rescueActionFromHeardText function, specifically designed to handle possible errors in the backend speech recognition. For example, when a child says "Open the intelligent object recognition system", the backend might misinterpret it as "Open the intelligent food system" or even "Open the intelligent food system", and the front end will automatically correct and execute the correct operation. This function covers dozens of common misrecognition situations, significantly reducing the frustration of children due to inaccurate pronunciation.

Wake-up word continuous listening (new feature): The final version has added the startWakeListening function, which uses the native Web Speech API of the browser to achieve continuous listening. Children no longer need to click any buttons; they just need to say the "stars" wake-up word, and the assistant will automatically activate and wait for instructions. This is a crucial leap from "manual triggering" to "touch-free interaction".

Daily Chat Function (New Feature): The final version has added the "callXingchenChat" function, specifically designed to handle non-educational open-ended conversations. Children can ask questions like "Tell me a story about stars",

"What's the weather like today", "Why are you called 'xingchen' ", etc. The assistant will respond through the /api/xingchen-chat interface. This makes the Xingchen Assistant no longer just a "learning tool", but a "companion" that can be accompanied and chatted with.

Intelligent Object Recognition System:

From "Single Camera + YOLO" to "YOLO + Coze Dual Mode" - Major Changes

The second version of the intelligent object recognition system relies on camera invocation + YOLO backend detection. The function is available but has two obvious issues: one is that the YOLO backend may be unstable or not started, and the other is that the recognition results only have pop-up windows and TTS reading, without subsequent learning and extension.

The final version has carried out a major reconfiguration of this module:

Camera invocation and real-time detection are fully retained: The camera invocation logic and the /detect interface invocation logic in the original camera.html have been completely transferred to the final version of panel-camera.

The result recognition pop-up window and TTS are completely retained: The objectModal pop-up window and the function of automatically reading out the recognition results remain unchanged.

Coze platform integration (new features): The final version has added iframe embedding for the Coze platform, with the address being <https://h4w79jmkxf.coze.site/>. This is an independent object recognition module developed by the team, offering an alternative object recognition experience. Additionally, considering that the Coze website might prohibit iframe embedding, the final version also provides "new tab opening" as a fallback option to ensure that users can use it in any situation.

Object Recognition Reporting Interface (New Feature): The final version has added the /api/object-detection/report interface. This enables the front end to report the object information (including labels and confidence levels) recognized by the camera to the back end, which is used for data statistics and learning analysis.

Object Recognition Learning Card Generation (New Feature): The final version has added the /api/object-learning/from-detection interface. When a child recognizes an object, they can request the backend to generate a "learning card", which includes a child-friendly description of the object, key words, and readable content, etc. This makes the object recognition process no longer a one-time "look and forget", but

instead transforms into an educational resource that can be repeatedly learned and accumulated over a long period of time.

UI/UX and Visual Design

The overall style has shifted to a dark cosmic theme: The main color is deep blue-purple-black, complemented by a complex background of stars, galaxies, and nebulae, creating an immersive sense of the universe. This style not only better aligns with the brand image of "Stars", but also is more easily recognizable for visually impaired and low-vision children, as the dark background combined with high-contrast bright text is easier to distinguish.

The Hero section on the homepage has been upgraded from simple text descriptions to a professional product showcase area: it includes gradient text titles, tag clouds, action buttons, and scrolling banners (Marquee). The scrolling banners continuously play brand slogans such as "Become the brightest star in the dark world" and "Listen to the world. Learn with starlight.", enhancing the product's dissemination power and memory points.

New design inspiration cards: The final version adds 3 "Inspo Cards" to the homepage, each explaining the source of one of the design concepts (galactic landing, dreamy nebula, advanced product UI). This detail has no direct functional impact, but in presentation or demonstration scenarios, it can clearly show the design thinking process to the judges.

The dynamic background has evolved from simple gradients to a complete cosmic scene: The final version generates complex dynamic backgrounds such as galactic bands, purple-blue pink nebulae, glass-like light balls, star particles, and meteors using CSS. These effects are not images but are generated purely by code, with good performance and not distorted due to changes in resolution.

The birth of the star fairy from nothing: This is the most eye-catching innovation in the visual aspect of the final version. The final version adds a complete interactive star fairy: The fairy can be dragged to any position on the screen; the fairy has 9 emotional states (idle, listening, thinking, happy, confused, sleepy, etc.), each state corresponding to different eye shapes, mouth curves and animation effects; the fairy will blink regularly, blinking randomly every 2-4 seconds; the fairy will pop up bubble text to answer or provide feedback in the form of bubbles; there are floating halos and two surrounding stars around the fairy. This fairy is not only a visual decoration, but also the "emotional core" of the entire product - it makes children feel that they are not using a tool, but interacting with a friend.

Custom icon support: The final version replaces the simple emoji logo in the second version with logo-img, which supports custom images. Users can use brand icons exported from tools like Canva to enhance the professionalism and brand recognition of the product.

Accessibility optimization has been significantly enhanced: The final version has added more semantic attributes such as aria-label, role, and aria-hidden. Combined with high-contrast text color schemes and more comprehensive TTS support, it enables screen reader users to better understand and use the page.

Responsive design is more comprehensive: The final version has been specially optimized for mobile device compatibility, including the size scaling of the galaxy band on narrow screens, and the grid layout of the navigation menu, ensuring a good user experience on both mobile phones and tablets.